

```
import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"^\jishnu",string )
print(match1)

import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"com$",string )
print(match1)
```

As shown in these two examples, the beginning and the end are being rightly detected (Figure 4).

The next character is '|' which signifies 'or'. The characters present before or after '|' will be detected as shown in Figure 5.

```
import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"j|s",string )
print(match1)

string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"saurav|penny",string )
print(match1)
```

The next character is the '?'. This verifies whether the given regex is there in the given string. Let us check it out using the following lines of code.

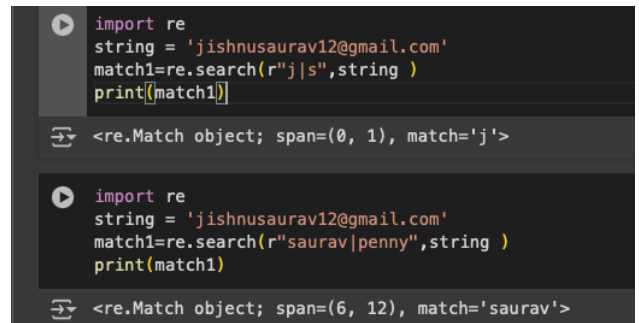
```
import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"a?",string )
print(match1)
```

We will get the index as 'nothing' because it is not there, but we will still get it and it will be recognised as shown in Figure 6.

We have explored the functionality of *re.search* that will help us search for a given regular expression pattern in a string. Let us now explore another important functionality of the 're' library, which is 'findall'. We can understand the remaining special characters of regular expressions through the *findall* functionality. All the special characters are applicable to all the functions in the *re* library.

The next character we are going to look at is '*'. The '*' implies that the pattern will be recognised if the regex is present zero or more times. The following code can be used to test this.

```
text = "jishnu, jiish, Raj"
matches = re.findall(r"ji*", text)
print(matches)
```



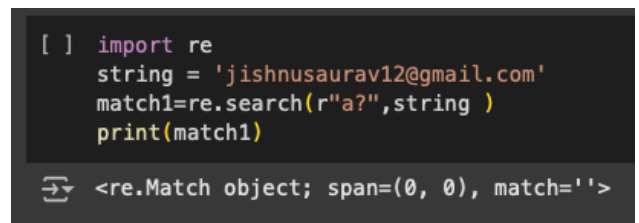
```
import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"j|s",string )
print(match1)

<re.Match object; span=(0, 1), match='j'>

import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"saurav|penny",string )
print(match1)

<re.Match object; span=(6, 12), match='saurav'>
```

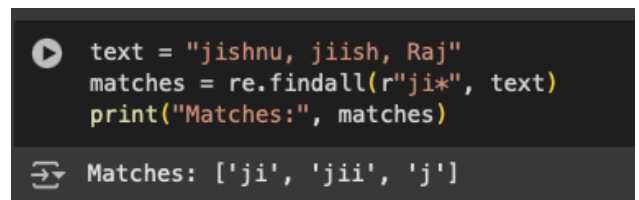
Figure 5: Functionality of 'or'



```
[ ] import re
string = 'jishnusaaurav12@gmail.com'
match1=re.search(r"a?",string )
print(match1)

<re.Match object; span=(0, 0), match=''>
```

Figure 6: Functionality of '?'



```
text = "jishnu, jiish, Raj"
matches = re.findall(r"ji*", text)
print("Matches:", matches)

Matches: ['ji', 'jii', 'j']
```

Figure 7: Functionality of '*'

As we can see in Figure 7, all the strings with 'j' followed by zero or more 'i's are detected and listed.

The next character is the '+'. The '+' implies that the pattern will be recognised if the regex is present one or more times. The same code as in the previous example can be used to test this.

```
text = "jishnu, jiish, Raj"
matches = re.findall(r"ji+", text)
print(matches)
```

As we can see in Figure 8, all the strings with 'j' followed by only one or more 'i's are detected and listed.

We now look at the character '{}', which is used to specify the count of the repeating regular expression that we want to verify.

This can be tested using the code given below. As we can see in Figure 9, only the words that have 'R' with three 'a's are being listed out.

```
text = "Raj Raaj Raaaj Raaaaj"
matches = re.findall(r"Ra{3}", text)
print("Matches:", matches)
```